NIDERSANA BALAJI

Software Engineering & Network Security Intern
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Computer Science Engineering (AI/ML) | Sona College of Technology

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PROFESSIONAL SUMMARY

Passionate Computer Science Engineering student with strong fundamentals in networking, security technologies, and software development. Demonstrated expertise in TCP/IP networking, cloud infrastructure, and security frameworks with hands-on experience in AWS, Docker, Kubernetes, and network debugging tools. Proven track record in competitive programming (85+ LeetCode problems solved) and AI/ML projects with optimization-focused development approach.

TECHNICAL SKILLS

Programming & Scripting Languages

- Python (Expert) – Performance optimization, automation scripting
- Java, C, C++ (Intermediate) – System programming and application development
- SQL/MySQL – Database design and query optimization
- Bash Scripting – System automation and deployment scripts

Networking & Infrastructure

- TCP/IP Networking Fundamentals – Routing protocols, switching concepts, subnetting
- Network Hardware – Routers, switches configuration and troubleshooting
- Network Debugging Tools – Wireshark, ping, traceroute, netstat, tcpdump
- DNS Management – Route 53, DNS resolution, routing policies

Security Technologies

- Firewall Configuration – Security groups, NACLs, iptables
- VPN & IPsec – Site-to-site VPN, client VPN configurations
- TLS/SSL – Certificate management, encryption protocols
- IPS/IDS Concepts – Intrusion detection and prevention systems

- Threat Analysis – Security vectors, vulnerability assessment

Cloud & DevOps

- **Amazon Web Services (AWS)**
    - EC2: Instance management, auto-scaling, load balancers
    - VPC: Subnets, route tables, security groups, NAT gateways
    - S3: Storage policies, lifecycle management, versioning
    - RDS: Database provisioning, backup strategies, monitoring
    - IAM: User/role management, security policies, least privilege
    - Lambda: Serverless computing, event-driven architectures

Infrastructure as Code & Automation

- Terraform – Infrastructure provisioning, state management, modules
- Ansible – Configuration management, playbooks, roles, ad-hoc commands
- Docker – Containerization, Dockerfile optimization, multi-stage builds
- Kubernetes – Pod orchestration, Services, Deployments, ConfigMaps, Secrets, Ingress
- CI/CD Pipelines – GitHub Actions, Jenkins, automated testing

Development & Testing

- Version Control – Git, GitHub, branching strategies, code reviews

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EDUCATION

Bachelor of Computer Science Engineering (AI/ML)
Sona College of Technology | 2023–2027
CGPA: **9.25/10**

Higher Secondary Education
Cluny Matriculation Higher Secondary School | 2021–2023
Percentage: **93%**

PROJECTS & ACHIEVEMENTS

- **GitTalk Agent – Repository Intelligence System**

- Enhanced GitHub code comprehension by 70% through RAG pipeline implementation
- Developed efficient caching mechanisms reducing API calls by 80%
- Implemented comprehensive error handling and logging systems

- **IoT Security Project – Women Safety IoT Device** *(First Prize, Intra-College Techathon)*
  - Integrated SOS, GPS, temperature, and heartbeat sensors
  - Implemented geofencing with real-time notification system
  - Developed secure communication protocols for nominee tracking

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COMPETITIVE ACHIEVEMENTS

Hackathons & Competitions

- **Smart India Hackathon 2024 – Waitlisted for Finals (7th among 500 teams)**
  - AI-Driven Gamified Freelancing Platform with secure payment gateway integration
- **Inter-College Hackathon – First Prize**
  - Enhanced AI freelancing platform with advanced security features
- **Startup Mahakumbh – FemOva.ai Presenter**
  - AI-powered wearable for PCOD/PCOS detection with IoT security implementation

Academic Excellence

- Academic Achievement Award – Three consecutive semesters (1st, 3rd, 1st rank)
- Speakers Forum – won 1st prize for two consecutive times followed by a 2nd prize
- Event Organization – Speakers Forum Event Coordinator

Technical Proficiency

- **LeetCode**: 85+ problems solved (Data Structures & Algorithms)
- **Networking Fundamentals**: Hands-on experience with routing protocols and switching

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ADDITIONAL TECHNICAL EXPERTISE

AI/ML & Automation Frameworks

- Agentic AI: LangGraph, LangChain, CrewAI, Agno
- MLOps: MLflow, Airflow, DVC, DagHub
- Data Analysis: PyTorch, Pandas, NumPy, Matplotlib, Seaborn

CORE COMPETENCIES

- **Technical Problem Solving** – Creative debugging and troubleshooting across network, security, and application layers
- **Performance Optimization** – Strong focus on code efficiency, network optimization, and system performance tuning
- **Team Collaboration** – Proven experience in cross-functional teams and stakeholder management
- **Continuous Learning** – Actively exploring emerging technologies in networking, security, and cloud computing
- **Communication Skills** – Effective technical communication in English, Telugu, and Tamil

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LANGUAGES

- English (Professional)
- Telugu (Native)
- Tamil (Fluent)
